CHAPTER I
INTRODUCTION

Rice is one of the most important cereal crops in the world and serves as the staple food for more than half of the global population (FAO, 2020). In many Asian countries, rice production plays an important role in food security, employment, and economic development. Myanmar is also recognized as one of the major rice producing countries in Southeast Asia, where rice has long been an important agricultural commodity for both domestic consumption and export activities (Juliano, 1993). Because rice is closely associated with consumer preference and market value, maintaining rice quality has become increasingly important in agricultural production and commercial trade.

Rice quality generally refers to the overall characteristics and attributes of rice grains that determine their market acceptability and economic value (Juliano, 1993). Consumers and traders commonly evaluate rice quality based on appearance characteristics such as grain size, grain shape, color, transparency, chalkiness, and visible defects. Among these characteristics, appearance quality is considered one of the most important factors because rice is usually judged visually before purchase (Verma, 2012). Rice grains with uniform shape, attractive appearance, and fewer defects are generally considered higher quality rice, whereas rice samples containing discoloration, chalkiness, or damaged grains are often categorized as lower quality rice (Rahman et al., 2020).

In international rice markets, rice quality has become increasingly important because importing countries require consistent quality standards and reliable grading systems. High quality rice generally receives higher market prices and stronger consumer acceptance, while poor quality rice may reduce commercial value and export competitiveness (FAO, 2020). Therefore, accurate and reliable rice quality evaluation is essential not only for consumers and traders but also for rice producers and exporters.

Traditionally, rice quality evaluation has been performed through manual inspection by experienced graders. In conventional grading practices, rice grains are visually examined to identify grain dimensions, discoloration, chalkiness, and other appearance defects. Although manual grading methods have been widely practiced for many years, they are often subjective and inconsistent because the evaluation process

depends heavily on human judgment and experience (Sakulsingharoj et al., 2010). Variations in lighting conditions, fatigue, and individual interpretation may also affect grading consistency and reliability. As a result, concerns regarding objectivity and accuracy in traditional rice grading systems have increased.

With the advancement of agricultural technology, digital image processing has become increasingly important in agricultural quality assessment. Digital image processing techniques allow image information to be transformed into measurable numerical data that can be analyzed objectively (Gonzalez & Woods, 2018). Previous studies reported that image-based rice quality evaluation systems improve grading consistency, reduce human error, and provide more efficient quality assessment procedures compared with traditional manual inspection methods (Ramos et al., 2018). These technological developments have encouraged researchers to investigate rice quality indicators derived from digital images.

1.1 Rationale of the Study

Rice quality assessment has become increasingly important because rice quality directly influences market value, export competitiveness, consumer preference, and food quality standards (FAO, 2020). In rice producing countries such as Myanmar, accurate rice grading systems are essential for supporting agricultural trade and maintaining product quality consistency. Since consumers usually evaluate rice visually before purchase, appearance characteristics such as grain size, grain shape, color, and defects strongly influence rice marketability and commercial value (Juliano, 1993). Therefore, objective rice quality evaluation methods are necessary for improving rice grading accuracy and maintaining quality standards in rice production and marketing systems.

Traditional rice grading systems mainly depend on manual inspection conducted by trained experts or rice graders. In manual grading methods, rice grains are visually inspected to determine grain dimensions, chalkiness, discoloration, and impurities. However, previous studies reported that manual inspection methods are associated with several limitations including subjectivity, inconsistency, human error, and time consumption (Sakulsingharoj et al., 2010). Different graders may produce different classification results because visual inspection is highly dependent on personal experience and environmental conditions. In addition, manual grading processes become inefficient when large quantities of rice samples need to be evaluated within

limited time periods. Consequently, automated and statistically reliable rice quality assessment systems have become increasingly necessary.

Advancements in digital image processing technologies have provided effective solutions for agricultural quality assessment problems (Gonzalez & Woods, 2018). Digital image processing techniques enable researchers to extract measurable image features from rice grain images objectively and consistently. Compared with manual inspection methods, image-based quality evaluation systems provide faster processing speed, greater accuracy, and improved grading consistency (Ramos et al., 2018). Therefore, digital image processing has become an important research area in agricultural engineering, food quality analysis, and computer vision applications.

Previous empirical studies demonstrated that geometric indicators significantly contribute to rice quality evaluation systems. Geometric indicators such as area, length, width, and aspect ratio provide quantitative information regarding grain dimensions and shape characteristics (Li et al., 2019). Rice grains with long, slender, and uniform shapes are generally associated with premium quality rice categories, whereas irregular or broken grains are usually classified into lower quality grades (Rahman et al., 2020). Therefore, geometric indicators are considered important variables in automated rice grading systems.

In addition to geometric characteristics, photometric indicators are also important for evaluating rice appearance quality and defect characteristics. Photometric indicators including RGB intensities, grayscale mean, variance, and uniformity index provide information related to grain color, texture, brightness, and surface consistency (Silva & Sonnadara, 2013). Previous studies reported that photometric variables significantly improve the identification of defective grains, discoloration, and texture irregularities in rice samples (Ponce-Garcia et al., 2017). Rice grains affected by yellow discoloration, chalkiness, or surface defects can be detected effectively through digital image analysis techniques.

Defect related indicators such as chalkiness rate, yellow rice rate, and spot rate are also important determinants of rice quality classification. Chalkiness negatively affects grain transparency and milling quality because chalky grains are more likely to break during milling processes (Kuo et al., 2016). Similarly, yellow rice defects and spotted grains reduce rice appearance quality and consumer acceptance (Zhang et al., 2021). Previous studies indicated that rice samples containing lower defect rates are generally classified into higher quality categories (Aung & Win, 2022). Therefore,

defect related indicators provide essential information for rice quality evaluation and automated grading systems.

Statistical analysis techniques also play an important role in rice quality classification studies. Multinomial logistic regression is widely used when the dependent variable consists of more than two categorical outcomes (Hosmer & Lemeshow, 2000). Previous empirical studies reported that multinomial logistic regression effectively analyzes the relationships between image derived rice quality indicators and rice grade categories (Hlaing et al., 2023). The technique allows researchers to evaluate how individual quality indicators influence the probability of rice samples being classified into different quality grades.

Although previous international studies have investigated rice quality analysis using digital image processing and statistical methods, limited studies have examined the combined effects of geometric indicators, photometric indicators, chalkiness rate, yellow rice rate, and spot rate on rice grade classification in Myanmar rice varieties. Moreover, limited studies have applied multinomial logistic regression analysis in the context of digital image-based rice quality evaluation in Myanmar. Therefore, conducting this study is important because it can provide statistically reliable and objective methods for rice quality assessment using digital image measurements.

This study was conducted to statistically analyze rice quality indicators derived from digital image measurements and examine their effects on rice grade classification using multinomial logistic regression analysis. The study is expected to contribute to the development of objective, efficient, and automated rice grading systems that can support rice quality control, agricultural research, and export competitiveness in Myanmar.

1.2 Objectives of the Study

The main objective of this study is to analyze rice quality indicators using digital image processing techniques and statistical analysis methods. The specific objectives of the study are as follows:

(i) To extract geometric and photometric features from digital rice grain images as quantitative rice quality indicators.

(ii) To conduct the comparative analysis of rice quality indicators based on rice grades.

(iii) To analyze the effect of rice quality indicators on rice grades.

1.3 Method of the Study

This study adopted a quantitative research design. Primary data were collected using a two-stage cluster sampling method. In the sampling procedure, a large rice sack was treated as the population. A “pyay tone” (traditional sampling bag) was selected as the first stage cluster, and a small cup was subsequently used to obtain the final subsample as the second stage cluster.

The sample consisted of three rice grades: Grade A, Grade B, and Grade C. A total of 100 digital images were collected from each rice grade, resulting in a total sample size of 300 rice grain images. The collected rice samples were scanned using a flatbed scanner under controlled lighting conditions to obtain high quality digital images for analysis.

After image acquisition, image preprocessing techniques including grayscale conversion, noise reduction, and thresholding were applied to improve image quality and facilitate segmentation. Image segmentation procedures were then performed to isolate individual rice grains from the background. Subsequently, feature extraction techniques were employed to obtain geometric indicators including area, length, width, and aspect ratio; photometric indicators including RGB intensities, grayscale mean, variance, and uniformity index; and defect related indicators including chalkiness rate, yellow rice rate, and spot rate.

After the feature extraction process, appropriate statistical techniques were applied for data analysis. Descriptive statistics including mean, standard deviation, minimum, and maximum values were used to summarize the rice quality indicators. One way Analysis of Variance (ANOVA) together with post hoc tests was employed to compare rice quality indicators across rice grades. Pearson’s correlation analysis was conducted to examine the relationships among the variables. Finally, multinomial logistic regression analysis was applied to examine the effects of rice quality indicators on rice grade classification.

1.4 Scope and Limitations of the Study

This study focused on the statistical analysis of rice quality indicators derived from digital image measurements. The study was limited to three rice grades categorized as Grade A, Grade B, and Grade C. A total of 300 rice grain images were analyzed, with 100 images collected from each rice grade. The study concentrated on image-based rice quality indicators, including geometric indicators (area, length, width,

and aspect ratio), photometric indicators (RGB intensities, grayscale mean, variance, and uniformity index), and defect related indicators (chalkiness rate, yellow rice rate, and spot rate). Multinomial logistic regression analysis was employed to examine the effects of these indicators on rice grade classification.

The study was limited to rice grain images obtained using a flatbed scanner under controlled lighting conditions. Therefore, the findings may vary under different imaging environments, camera systems, or lighting conditions. In addition, the study focused only on appearance related quality indicators derived from digital images and did not include chemical properties, cooking quality, aroma, taste, or nutritional characteristics of rice. The statistical analysis was also limited to the selected rice samples and may not fully represent all Myanmar rice varieties. Furthermore, the study applied multinomial logistic regression analysis as the primary classification method; therefore, the findings may differ when other machine learning or advanced classification techniques are used.

1.5 Organization of the Study

This study was organized into five chapters. Chapter I presented the introduction, rationale of the study, objectives of the study, scope and method of the study, and organization of the study. Chapter II reviewed the relevant literature related to rice quality assessment, digital image processing, rice quality indicators, previous empirical studies, and the conceptual framework of the study. Chapter III described the research methodology, including data collection procedures, variable measurements, and statistical analysis methods used in the study. Chapter IV presented the analysis and findings of the study. Finally, Chapter V provided the findings of the study, suggestions and recommendations, and needs for further studies.

CHAPTER II
LITERATURE REVIEW

This chapter presents a review of the relevant literature related to rice quality assessment, digital image processing, rice grading systems, image-based feature extraction techniques, and statistical analysis methods used in rice quality evaluation. The concepts of geometric and photometric indicators, chalkiness rate, yellow rice rate, and spot rate in rice quality analysis are also discussed. In addition, previous empirical studies related to digital image measurements and rice quality classification are reviewed in order to establish the theoretical and analytical foundation of the study. Finally, the conceptual framework of the study is explained in relation to the literature.

2.1 Rice and Rice Quality

Rice was recognized as one of the most important cereal crops in the world and was consumed as the staple food by more than half of the global population (FAO, 2020). It played a vital role in food security, nutrition, employment, and economic development, particularly in Asian countries. Myanmar was recognized as one of the major rice producing countries in Southeast Asia, and rice production was regarded as a significant component of the country's agricultural sector (Juliano, 1993). Owing to the importance of rice in both domestic consumption and international trade, maintaining rice quality had become increasingly essential (FAO, 2020).

Rice quality was defined as the overall characteristics and attributes of rice grains that determined their market value, consumer preference, and suitability for processing and consumption (Juliano, 1993). Rice quality was generally evaluated based on physical, chemical, cooking, milling, and appearance characteristics. Among these factors, physical appearance quality was considered one of the most important aspects because consumers and traders often judged rice quality visually before purchase (Verma, 2012).

According to Juliano (1993), rice quality included several physical attributes such as grain size, grain shape, color, transparency, chalkiness, and the presence of broken or defective grains. These visual characteristics influenced both market price and consumer acceptance. Long and slender rice grains with uniform color and fewer defects were generally considered higher quality rice (Li et al., 2019).

Traditionally, rice quality assessment was conducted through manual inspection by experts or graders. In traditional grading systems, rice grains were visually examined to determine characteristics such as grain length, grain width, chalkiness, discoloration, and impurities. Although manual inspection was widely practiced, it had several limitations because human evaluation could be subjective, time consuming, inconsistent, and affected by fatigue or environmental conditions (Sakulsingharoj et al., 2010).

As agricultural technology advanced, digital image processing and statistical analysis techniques were increasingly applied to rice quality assessment (Gonzalez & Woods, 2018). These techniques provided more objective, accurate, and efficient methods for evaluating rice quality. Image based analysis allowed rice grains to be measured quantitatively using geometric and photometric indicators derived from digital images (Ramos et al., 2018).

Rice quality became increasingly important in international trade because export standards required consistency and quality assurance. Countries importing rice products established strict quality standards regarding grain appearance, purity, and defect rates (FAO, 2020). Therefore, accurate rice quality assessment systems were considered essential for maintaining competitiveness in global markets. Furthermore, rice quality directly influenced consumer satisfaction and economic value. Higher quality rice generally commanded higher market prices, whereas poor quality rice reduced consumer confidence and commercial value (Rahman et al., 2020). Consequently, the development of reliable and automated rice quality assessment methods became a major research interest in agricultural engineering, computer vision, and statistical analysis (Zhang et al., 2021).

2.2 Rice Grading Systems

Rice grading was defined as the process of classifying rice into different quality categories based on predetermined standards and quality indicators (FAO, 2020). Rice grading systems were used by governments, exporters, millers, and traders to ensure consistency, maintain product quality, and facilitate fair market pricing (Juliano, 1993).

Grading standards varied across countries depending on market requirements and national regulations. However, most grading systems considered similar quality indicators such as grain size, shape, color, chalkiness, damaged grains, impurities, and

moisture content (Verma, 2012). Higher grades were generally assigned to rice with better physical appearance, fewer defects, and greater uniformity (Li et al., 2019).

The Food and Agriculture Organization (FAO) reported that rice grading systems were important for ensuring quality control and improving international trade efficiency (FAO, 2020). In many countries, rice grades were categorized into classes such as Grade A, Grade B, and Grade C according to quality level. Grade A rice generally represented high quality rice with long and uniform grains, minimal chalkiness, low defect rates, and attractive appearance (Rahman et al., 2020). Grade B rice contained moderate levels of defects or irregularities, whereas Grade C rice included higher levels of damaged grains, discoloration, or chalkiness (Ramos et al., 2018).

Traditional rice grading systems relied heavily on manual inspection and expert judgment. Although experienced graders provided useful evaluations, manual grading methods often produced inconsistent results because of human subjectivity (Sakulsingharoj et al., 2010). Technological advancements encouraged the development of automated rice grading systems using machine vision and image processing techniques (Gonzalez & Woods, 2018). Automated systems analyzed rice grain characteristics objectively and rapidly using digital images, thereby improving accuracy, consistency, and efficiency (Silva & Sonnadara, 2013).

Digital image-based rice grading systems generally involved image acquisition, preprocessing, segmentation, feature extraction, statistical analysis, and classification procedures (Gonzalez & Woods, 2018). Various machine learning and statistical models were applied to classify rice grades based on extracted image features (Ramos et al., 2018). Researchers increasingly focused on developing intelligent grading systems capable of identifying rice quality categories automatically (Zhang et al., 2021).

Rice grading was considered important not only for commercial purposes but also for agricultural research and quality improvement. Accurate grading systems helped rice producers and processors identify quality problems and improve production processes (Rahman et al., 2020). Therefore, rice grading systems played an important role in maintaining rice quality standards, supporting fair pricing mechanisms, and enhancing market competitiveness (FAO, 2020).

2.3 Digital Image Processing

Digital image processing was defined as the use of computer algorithms and computational techniques to analyze, manipulate, and extract information from digital images (Gonzalez & Woods, 2018). Image processing technology became increasingly important in agricultural applications because it enabled objective and automated analysis of agricultural products (Verma, 2012).

According to Gonzalez and Woods (2018), digital image processing involved transforming visual information into numerical data that could be analyzed computationally. In agricultural quality assessment, image processing techniques were commonly used to evaluate fruits, vegetables, grains, and other food products (Silva & Sonnadara, 2013). A digital image processing system generally consisted of several stages: image acquisition, preprocessing, segmentation, feature extraction, and classification (Ramos et al., 2018). These stages enabled researchers to convert raw images into meaningful quantitative information (Li et al., 2019).

Previous studies reported that digital image processing techniques improved the accuracy, consistency, and efficiency of agricultural quality assessment systems (Zhang et al., 2021). In rice quality analysis, digital image processing methods were widely applied to measure grain size, grain shape, color characteristics, chalkiness, and surface defects automatically (Sakulsingharoj et al., 2010). Automated image analysis systems reduced human subjectivity and provided more reliable quality evaluation results compared with traditional manual inspection methods (Rahman et al., 2020).

2.3.1 Image Acquisition

Image acquisition was considered the first stage of digital image processing. It involved capturing images using digital cameras, scanners, or imaging sensors (Gonzalez & Woods, 2018). In rice quality assessment, rice grain images were usually captured under controlled lighting and background conditions to ensure image consistency (Silva & Sonnadara, 2013).

Proper image acquisition was regarded as important because image quality significantly influenced the accuracy of subsequent analysis (Ramos et al., 2018). Factors such as lighting intensity, camera resolution, image orientation, and background color affected the visibility and measurement of rice grain features (Verma, 2012). Researchers commonly used white or black backgrounds to improve image contrast and facilitate grain segmentation (Sakulsingharoj et al., 2010). Uniform lighting conditions

were also considered essential for reducing shadows and reflections during image capture (Li et al., 2019).

Previous studies reported that poor image acquisition conditions reduced the accuracy of feature extraction and classification results in rice quality analysis systems (Rahman et al., 2020). Therefore, standardized image acquisition procedures were widely applied in digital image-based rice quality assessment studies (Zhang et al., 2021).

2.3.2 Image Preprocessing

Image preprocessing was defined as a set of techniques used to improve image quality before feature extraction (Gonzalez & Woods, 2018). Preprocessing techniques were applied to remove noise, enhance image clarity, and standardize image characteristics for subsequent analysis (Ramos et al., 2018). Common preprocessing techniques included image resizing, filtering, grayscale conversion, thresholding, and contrast enhancement (Verma, 2012). These techniques improved the accuracy and reliability of rice grain detection and feature measurement (Li et al., 2019).

Noise reduction techniques such as median filtering and Gaussian filtering were frequently used to eliminate unwanted disturbances and image noise (Silva & Sonnadara, 2013). Thresholding techniques were applied to separate rice grains from the image background and to facilitate segmentation processes (Sakulsingharoj et al., 2010). Previous studies reported that effective preprocessing techniques significantly improved the performance of image-based rice quality assessment systems (Rahman et al., 2020). Zhang et al. (2021) also found that preprocessing methods enhanced classification accuracy by improving image consistency and reducing irrelevant image variations.

2.3.3 Image Segmentation

Image segmentation was defined as the process of dividing an image into meaningful regions or objects (Gonzalez & Woods, 2018). In rice quality analysis, segmentation techniques were used to isolate individual rice grains from the image background (Sakulsingharoj et al., 2010). Various segmentation techniques including threshold-based segmentation, edge detection, region growing, and clustering algorithms were applied in digital image processing systems (Verma, 2012). These

techniques enabled researchers to identify rice grain boundaries and extract grain regions accurately (Li et al., 2019).

Accurate segmentation was considered important because segmentation errors directly affected feature extraction and classification results (Ramos et al., 2018). Previous studies reported that improper segmentation reduced the accuracy of rice quality assessment systems and produced unreliable measurements (Rahman et al., 2020). Binary image conversion was commonly used in rice image analysis, where rice grains were represented as white objects and the background was represented as black regions (Silva & Sonnadara, 2013). Zhang et al. (2021) reported that binary segmentation techniques improved the efficiency of automated rice grain detection and defect analysis systems.

2.3.4 Feature Extraction

Feature extraction was considered one of the most important stages in digital image processing because measurable characteristics were extracted from images to represent object properties quantitatively (Gonzalez & Woods, 2018). In rice quality analysis, extracted features generally included geometric features and photometric features (Ramos et al., 2018). These features provided information related to grain shape, size, color, texture, and overall appearance (Verma, 2012). Geometric features described physical grain dimensions such as length, width, area, and aspect ratio, whereas photometric features represented intensity and color properties derived from image pixels (Li et al., 2019).

Previous studies reported that feature extraction significantly contributed to the accuracy of rice quality classification systems (Sakulsingharoj et al., 2010). Silva & Sonnadara (2013) extracted color and texture related features from rice grain images and found that the extracted variables effectively differentiated rice quality categories. Feature extraction enabled researchers to convert visual information into numerical variables that could be analyzed statistically (Rahman et al., 2020). Zhang et al. (2021) also reported that properly extracted geometric and photometric features improved the performance of automated rice defect detection and classification systems.

2.3.5 Image Classification

Image classification was defined as the process of assigning images or objects into predefined categories based on extracted features (Gonzalez & Woods, 2018). In rice quality studies, classification methods were applied to categorize rice samples into different quality grades (Sakulsingharoj et al., 2010). Various classification methods including statistical models and machine learning algorithms were applied in agricultural image analysis systems (Verma, 2012). Previous studies reported that classification techniques improved the accuracy and efficiency of automated rice quality assessment systems (Ramos et al., 2018). Silva & Sonnadara (2013) applied image classification techniques using color and texture features extracted from rice grain images and found that classification performance significantly improved with appropriate feature selection. Similarly, Zhang et al. (2021) employed machine learning based classification methods for automated rice defect detection and reported high classification accuracy.

In the context of this study, multinomial logistic regression was employed as the classification method to predict rice grades (A, B, and C) based on extracted geometric and photometric indicators. Hosmer & Lemeshow (2000) stated that multinomial logistic regression was appropriate when the dependent variable contained more than two categorical outcomes. Hlaing et al. (2023) applied multinomial logistic regression analysis to examine the effects of rice quality indicators on rice grade classification. The study reported that geometric and photometric variables significantly influenced rice grade prediction. Automated classification systems provided several advantages such as faster processing speed, greater consistency, and reduced human error compared with traditional manual grading systems (Rahman et al., 2020). Therefore, image classification techniques were increasingly adopted in rice quality assessment and agricultural quality control systems (Li et al., 2019).

2.4 Rice Quality Indicators

Rice quality indicators were defined as measurable characteristics derived from digital image analysis and statistical measurements that were used to evaluate rice quality objectively (Ramos et al., 2018). Previous studies reported that image-based quality indicators improved the consistency and accuracy of rice grading systems compared with traditional manual inspection methods (Rahman et al., 2020). Researchers mainly focused on geometric indicators, photometric indicators, and defect

related indicators such as chalkiness rate, yellow rice rate, and spot rate when analyzing rice quality using digital image processing techniques (Sakulsingharoj et al., 2010). These variables were commonly extracted from rice grain images and statistically analyzed to determine their relationships with rice grades (Li et al., 2019).

Verma (2012) reported that image derived quality indicators provided more reliable and quantitative measurements for rice quality evaluation. Similarly, Silva & Sonnadara (2013) found that geometric and color related indicators significantly improved rice grain classification performance. Ramos et al. (2018) analyzed rice grading using geometric and photometric image descriptors and reported that image-based indicators significantly influenced rice grade classification accuracy. Zhang et al. (2021) also stated that defect related indicators such as discoloration, spots, and texture irregularities were important variables in automated rice quality assessment systems. Previous empirical studies consistently suggested that rice quality indicators derived from digital image measurements provided objective, efficient, and statistically reliable methods for evaluating rice quality categories (Hlaing et al., 2023).

2.5 Geometric Indicators

Geometric indicators were defined as the physical dimensions and shape characteristics of rice grains obtained from digital image measurements (Ramos et al., 2018). These indicators were widely used in rice quality analysis because grain size and shape characteristics were considered important determinants of rice appearance quality and market value (Juliano, 1993). Previous studies reported that geometric features significantly contributed to rice classification and quality evaluation systems (Verma, 2012). Common geometric indicators included grain area, grain length, grain width, perimeter, and aspect ratio, which were extracted from rice grain images using image processing techniques (Li et al., 2019).

Sakulsingharoj et al. (2010) conducted a study on rice grain classification using image processing and statistical analysis techniques. In their study, grain area, grain length, grain width, and aspect ratio were extracted from rice grain images. The researchers reported that geometric indicators effectively differentiated rice grades and rice varieties. The findings suggested that grain shape characteristics significantly improved classification performance. Verma (2012) investigated rice quality analysis using digital image processing methods. Shape based features including grain length, grain width, and grain area were extracted and statistically analyzed. The findings

indicated that geometric variables provided reliable and objective measurements for automated rice quality assessment systems.

Li et al. (2019) developed a machine vision system for rice appearance quality inspection. Grain dimensions and shape descriptors were measured using digital image processing techniques. The researchers found that geometric indicators significantly influenced rice grade classification accuracy. Long and slender rice grains with consistent dimensions were generally associated with higher quality grades. Rahman et al. (2020) conducted a comparative analysis of rice quality indicators using image-based measurements. Grain size and shape characteristics were compared among different rice quality categories. Significant differences in grain length, width, and aspect ratio were observed across rice grades. The study concluded that geometric indicators were effective variables for distinguishing rice quality categories.

Silva & Sonnadara (2013) also applied geometric feature analysis in rice grain classification studies and reported that shape related variables significantly improved the accuracy of automated rice grading systems. Similarly, Ramos et al. (2018) analyzed rice grading using geometric image descriptors and found that grain shape indicators contributed significantly to rice quality prediction models. Previous empirical studies consistently indicated that geometric indicators derived from digital image measurements provided objective, accurate, and statistically reliable information for rice quality evaluation and rice grade classification (Hlaing et al., 2023).

2.5.1 Area

Area was defined as the total pixel region occupied by a rice grain in a digital image (Gonzalez & Woods, 2018). Area measurements were commonly used to evaluate grain size and detect abnormal, damaged, or broken grains in rice quality assessment systems (Verma, 2012). In digital image analysis, grain area was calculated by counting the number of pixels representing the rice grain after image segmentation and binary image conversion processes were completed (Sakulsingharoj et al., 2010). Previous studies reported that grain area was an important geometric indicator for distinguishing rice quality categories because grain size uniformity strongly influenced rice appearance quality and market value (Juliano, 1993). Rice samples containing grains with consistent area distributions were generally associated with higher quality grades, whereas irregular area distributions often indicated the presence of broken or defective grains (Rahman et al., 2020).

Sakulsingharoj et al. (2010) measured grain area to classify rice samples using digital image processing techniques. The researchers reported that area values varied significantly among rice categories and rice varieties. Their findings suggested that grain area contributed significantly to automated rice classification performance. Similarly, Verma (2012) investigated rice quality analysis using digital image processing methods and found that grain area significantly contributed to rice grain identification and quality assessment. The study reported that area measurements provided objective and quantitative information for evaluating rice grain quality.

Li et al. (2019) also analyzed grain area in a machine vision system developed for rice appearance quality inspection. The researchers observed that rice samples with uniform grain area distributions were more likely to be classified into higher quality categories. The study further indicated that abnormal grain area measurements were frequently associated with broken grains and defective rice samples. Aung & Win (2022) applied image processing techniques to evaluate rice quality in Myanmar. Grain area measurements were extracted from binary rice images, and the findings showed that higher quality rice samples generally exhibited more uniform grain area distributions. The researchers concluded that grain area was an effective geometric indicator for rice quality evaluation in automated grading systems. Previous empirical studies consistently suggested that grain area measurements derived from digital image analysis provided reliable and statistically meaningful information for rice quality assessment and rice grade classification (Hlaing et al., 2023).

2.5.2 Length

Grain length was considered one of the most important rice quality indicators because consumers and traders generally preferred long and uniform rice grains (Juliano, 1993). Grain length significantly influenced rice appearance quality, market value, and consumer acceptance, particularly in international rice markets where long grain rice was often associated with premium quality standards (FAO, 2020). In digital image analysis, grain length was measured as the maximum distance between the two ends of a rice grain obtained from segmented rice images (Gonzalez & Woods, 2018). Previous studies reported that grain length was widely used as an important geometric indicator in automated rice quality assessment systems (Verma, 2012).

Li et al. (2019) analyzed grain length using machine vision techniques and reported that long grain rice samples were more frequently associated with premium

quality grades. The researchers found that grain length significantly improved rice grade classification accuracy in machine vision-based quality inspection systems. Rahman et al. (2020) conducted a comparative analysis of rice quality indicators using image-based measurements and reported that grain length differed significantly among rice quality categories. Their findings indicated that higher grade rice samples generally contained longer and more uniform grains compared with lower grade samples.

Sakulsingharoj et al. (2010) also extracted grain length measurements from rice grain images and reported that grain length significantly contributed to rice variety classification and rice quality differentiation. The study suggested that grain length provided objective and reliable information for automated rice grading systems. Similarly, Verma (2012) found that grain length measurements improved the performance of digital image-based rice quality evaluation systems. The researcher reported that grain length was one of the most effective variables for distinguishing high-quality rice from defective or low-quality rice samples.

Hlaing et al. (2023) statistically analyzed rice quality indicators using multinomial logistic regression analysis. Their findings suggested that grain length significantly influenced rice grade classification. The study reported that rice samples with longer grain measurements were more likely to be classified into higher quality categories. Previous empirical studies consistently indicated that grain length derived from digital image measurements provided accurate, objective, and statistically meaningful information for rice quality evaluation and automated rice classification systems (Ramos et al., 2018).

2.5.3 Width

Width referred to the shorter dimension of rice grains measured perpendicular to grain length (Gonzalez & Woods, 2018). Width measurements were frequently used together with grain length to evaluate grain proportion, grain shape, and overall rice appearance quality (Juliano, 1993). In digital image analysis, grain width was measured automatically from segmented rice grain images using image processing techniques (Verma, 2012). Previous studies reported that grain width was an important geometric indicator in rice quality assessment because width measurements significantly influenced grain shape classification and rice appearance evaluation (Li et al., 2019). Rice grains with consistent width measurements were generally associated with higher quality grades, whereas irregular or abnormal width values were often related to

damaged, immature, or broken grains (Rahman et al., 2020). Verma (2012) investigated rice quality analysis using digital image processing methods and reported that grain width measurements improved the accuracy of automated rice classification systems. The study suggested that width measurements provided reliable and objective information for identifying rice grain characteristics.

Li et al. (2019) also analyzed grain width in a machine vision system developed for rice appearance quality inspection. The researchers found that grain width significantly contributed to rice appearance evaluation and classification accuracy. Their findings indicated that rice samples with uniform width distributions were more likely to be classified into higher quality categories. Sakulsingharoj et al. (2010) extracted width measurements from rice grain images and reported that grain width, together with grain length and aspect ratio, effectively differentiated rice varieties and rice grades. Similarly, Ramos et al. (2018) found that width measurements significantly contributed to image-based rice grading models.

Rahman et al. (2020) compared geometric indicators among different rice quality categories and observed significant differences in grain width values across rice grades. The study concluded that abnormal grain width values were frequently associated with damaged or low-quality rice grains. Previous empirical studies consistently suggested that grain width derived from digital image measurements provided statistically reliable and objective information for rice quality evaluation and automated rice grading systems (Hlaing et al., 2023).

2.5.4 Aspect Ratio

Aspect ratio was defined as the ratio of grain length to grain width and was widely used to evaluate rice grain shape characteristics (Gonzalez & Woods, 2018). Aspect ratio was considered an important geometric indicator because it reflected the slenderness and proportional shape of rice grains (Juliano, 1993). In rice quality evaluation, grains with higher aspect ratio values were generally regarded as longer and more-slender, characteristics that were often preferred by consumers and traders (Rahman et al., 2020). Previous studies reported that aspect ratio significantly contributed to rice classification and rice appearance evaluation systems (Verma, 2012). Aspect ratio measurements were commonly extracted from segmented rice grain images using digital image processing techniques and were statistically analyzed to distinguish rice varieties and quality grades (Ramos et al., 2018).

Sakulsingharoj et al. (2010) conducted rice grain classification studies using image processing and statistical analysis techniques. The researchers reported that aspect ratio was one of the most discriminative geometric indicators in rice classification systems. Their findings indicated that slender grains with higher aspect ratio values were generally associated with better quality rice and premium rice grades. Rahman et al. (2020) conducted a comparative analysis of rice quality indicators using image-based measurements and observed that aspect ratio significantly differed among rice grades. Higher quality rice samples generally exhibited more uniform and higher aspect ratio values compared with lower quality samples.

Li et al. (2019) also analyzed aspect ratio using machine vision techniques for rice appearance quality inspection. The study reported that aspect ratio significantly improved classification accuracy in automated rice grading systems because grain shape characteristics strongly influenced rice appearance quality. Similarly, Verma (2012) found that aspect ratio measurements provided reliable and objective information for evaluating rice grain shape and identifying defective or irregular grains. The researcher concluded that aspect ratio was an effective variable for automated rice quality assessment.

Hlaing et al. (2023) statistically analyzed rice quality indicators using multinomial logistic regression analysis and found that aspect ratio significantly affected rice grade classification. The findings suggested that rice samples with higher aspect ratio values were more likely to be classified into higher quality categories. Previous empirical studies consistently indicated that aspect ratio derived from digital image measurements provided statistically meaningful and reliable information for rice quality evaluation and automated rice grading systems (Zhang et al., 2021).

2.6 Photometric Indicators

Photometric indicators were defined as image features related to brightness, color intensity, and texture characteristics derived from digital images (Gonzalez & Woods, 2018). These indicators were widely used in rice quality analysis because color and texture characteristics significantly influenced rice appearance quality and defect detection (Ramos et al., 2018). Previous studies reported that photometric variables were important for detecting discoloration, defective grains, surface irregularities, and overall rice appearance quality (Silva & Sonnadara, 2013). Common photometric indicators included RGB color intensities, grayscale mean, variance, and uniformity

index, which were extracted from rice grain images using digital image processing techniques (Verma, 2012).

Silva & Sonnadara (2013) conducted rice grain quality analysis using image processing techniques. RGB color intensities and grayscale features were extracted from rice grain images and statistically analyzed. The researchers reported that photometric indicators significantly improved rice classification performance and enhanced the identification of defective rice grains. Ponce Garcia et al. (2017) applied texture analysis techniques for rice defect detection. Variance and uniformity measurements were analyzed statistically, and the findings indicated that photometric features significantly contributed to defect identification and texture evaluation. The study suggested that texture-based indicators improved the accuracy of automated rice quality assessment systems.

Ramos et al. (2018) analyzed rice grading using geometric and photometric image descriptors. Their results indicated that both color-based and texture-based indicators significantly influenced rice grade classification. The researchers reported that photometric variables improved the performance of image-based rice grading models by providing detailed information related to grain brightness and surface characteristics. Li et al. (2019) also applied photometric feature analysis in machine-vision based rice appearance inspection systems. The study found that color intensity and texture related variables significantly contributed to the identification of rice quality categories and defective grains.

Similarly, Zhang et al. (2021) investigated automated rice defect detection using image processing and machine learning techniques. The researchers reported that photometric indicators effectively detected discoloration, texture irregularities, and abnormal grain surfaces in rice samples. Previous empirical studies consistently indicated that photometric indicators derived from digital image measurements provided objective, reliable, and statistically meaningful information for rice quality evaluation and automated rice grading systems (Hlaing et al., 2023).

2.6.1 RGB Intensities

RGB intensities represented the red, green, and blue color components extracted from digital images (Gonzalez & Woods, 2018). These color components were commonly used to analyze grain color, discoloration, and appearance characteristics in rice quality assessment systems (Verma, 2012). RGB intensity measurements provided

quantitative information related to color variation and surface quality of rice grains (Ramos et al., 2018). Silva & Sonnadara (2013) extracted RGB intensity values from rice grain images and reported that color information effectively distinguished defective and non-defective grains. The study indicated that RGB features significantly improved automated rice classification performance.

Ramos et al. (2018) also analyzed rice grading using color-based image descriptors and found that RGB intensity variables significantly contributed to rice quality classification. The researchers reported that color related indicators provided reliable information for identifying rice quality categories and defective grains. Zhang et al. (2021) investigated automated rice defect detection using image texture and color analysis methods. The study reported that RGB intensity variations were associated with rice discoloration, yellow grains, and surface defects. The researchers concluded that RGB color analysis improved the detection accuracy of defective rice samples.

Similarly, Li et al. (2019) applied RGB intensity analysis in machine vision-based rice quality inspection systems and found that color intensity variables significantly influenced rice appearance evaluation and rice grade prediction. Previous empirical studies consistently indicated that RGB intensity measurements derived from digital image analysis provided objective and statistically reliable information for rice quality evaluation and automated rice grading systems (Hlaing et al., 2023).

2.6.2 Grayscale Mean

Grayscale mean referred to the average intensity value obtained after converting color images into grayscale images (Gonzalez & Woods, 2018). Grayscale conversion simplified image analysis by reducing color information into intensity levels ranging from black to white (Verma, 2012). In rice quality assessment, grayscale mean values were commonly used to evaluate grain brightness, surface appearance, and texture characteristics (Ramos et al., 2018). Previous studies reported that grayscale mean measurements significantly contributed to the detection of rice defects and appearance irregularities (Ponce Garcia et al., 2017). Higher grayscale mean values were generally associated with brighter and more visually uniform rice grains, whereas lower grayscale values often indicated defective or discolored grains (Kuo et al., 2016).

Kuo et al. (2016) used grayscale intensity analysis to detect chalkiness in rice grains. The researchers reported that brighter grains produced higher grayscale mean values, while defective grains exhibited lower grayscale intensity levels. The study

concluded that grayscale analysis provided effective measurements for identifying chalky and defective rice grains. Ponce Garcia et al. (2017) also applied grayscale measurements in rice texture analysis and found that grayscale mean values significantly contributed to rice appearance evaluation and defect identification. Their findings suggested that grayscale features improved the performance of automated rice quality assessment systems.

Similarly, Zhang et al. (2021) analyzed grayscale intensity features in automated rice defect detection models and reported that grayscale mean measurements effectively distinguished normal grains from defective grains with discoloration or surface irregularities. Previous empirical studies consistently indicated that grayscale mean values derived from digital image analysis provided objective and statistically meaningful information for rice quality evaluation and automated rice grading systems (Hlaing et al., 2023).

2.6.3 Variance

Variance was defined as a statistical measurement used to evaluate the variability of pixel intensity values within rice grain images (Gonzalez & Woods, 2018). In digital image analysis, variance measurements were commonly applied to assess texture irregularities, surface roughness, and intensity distribution characteristics of rice grains (Ponce Garcia et al., 2017). Previous studies reported that variance values significantly contributed to rice defect detection because defective grains generally exhibited greater intensity fluctuations and irregular surface textures compared with normal grains (Ramos et al., 2018). Higher variance values were often associated with chalkiness, discoloration, cracks, and surface defects in rice grain images (Verma, 2012).

Ponce Garcia et al. (2017) analyzed variance measurements to identify rice grain defects using texture analysis techniques. The researchers reported that defective grains exhibited higher intensity variance values because of irregular surface texture and inconsistent brightness distribution. Their findings suggested that variance measurements improved the accuracy of automated rice defect identification systems. Zhang et al. (2021) also investigated automated rice defect detection using image texture and statistical analysis methods. The study found that variance measurements significantly improved defect detection accuracy in rice quality classification systems.

Rice grains with abnormal texture patterns and surface irregularities generally produced higher variance values.

Similarly, Silva & Sonnadara (2013) applied statistical texture analysis in rice quality assessment and reported that variance related features effectively differentiated normal grains from defective grains. The study concluded that texture variability measurements provided useful information for automated rice grading systems. Previous empirical studies consistently indicated that variance measurements derived from digital image analysis provided objective and statistically reliable information for evaluating rice grain texture, detecting defects, and improving rice quality classification performance (Hlaing et al., 2023).

2.6.4 Uniformity Index

Uniformity index was defined as a measurement used to evaluate the consistency of pixel intensity distribution within rice grain images (Gonzalez & Woods, 2018). In digital image analysis, uniformity measurements were commonly applied to assess the smoothness, consistency, and visual regularity of rice grain surfaces (Ponce Garcia et al., 2017). Previous studies reported that rice grains with more consistent surface appearance generally produced higher uniformity values, whereas defective or irregular grains produced lower uniformity measurements (Ramos et al., 2018). Uniformity index was considered an important photometric indicator because it provided quantitative information related to rice appearance quality and texture consistency (Verma, 2012).

Ramos et al. (2018) analyzed rice grading using geometric and photometric image descriptors and reported that rice grains with higher quality appearance generally produced higher uniformity values. The researchers suggested that uniformity analysis improved the identification of visually consistent rice samples and enhanced automated rice grading performance. Ponce Garcia et al. (2017) also applied texture analysis techniques in rice defect detection studies and found that lower uniformity values were associated with surface irregularities, defective grains, and inconsistent texture patterns. Their findings indicated that uniformity measurements significantly contributed to rice quality evaluation and defect identification.

Similarly, Zhang et al. (2021) investigated automated rice defect detection using image texture analysis methods and reported that uniformity related features effectively differentiated normal rice grains from defective grains with abnormal surface

characteristics. Silva & Sonnadara (2013) further reported that texture consistency measurements improved the accuracy of rice classification systems because higher quality rice samples generally exhibited more uniform surface characteristics and brightness distributions. In the conceptual framework of this study, the uniformity index was included as one of the important photometric indicators derived from digital images. Previous empirical studies consistently indicated that uniformity measurements provided objective and statistically reliable information for evaluating rice appearance quality and automated rice grading systems (Hlaing et al., 2023).

2.7 Chalkiness Rate

Chalkiness rate referred to the proportion of chalky regions or chalky grains present in rice samples (Juliano, 1993). Chalkiness was considered one of the most important rice quality defects because chalky grains were more susceptible to breakage during milling processes and were visually less attractive to consumers and traders (FAO, 2020). Rice samples with high chalkiness levels were generally associated with lower market value and reduced rice quality grades (Rahman et al., 2020). Previous studies reported that chalkiness significantly influenced rice appearance quality, milling quality, and consumer acceptance (Verma, 2012). Chalky grains usually exhibited opaque white regions caused by irregular starch granule arrangements inside the grain structure (Kuo et al., 2016). These chalky regions reduced grain transparency and weakened grain hardness, thereby increasing the probability of grain breakage during milling operations (Li et al., 2019).

Kuo et al. (2016) investigated chalkiness detection using digital image analysis techniques. Chalky regions were segmented from rice grain images and chalkiness levels were quantified statistically. The researchers reported that image processing methods effectively detected chalkiness defects and provided objective measurements for rice quality evaluation. Li et al. (2019) examined rice appearance quality using machine vision systems and reported that chalkiness significantly influenced rice grade classification. Their findings indicated that rice samples containing lower chalkiness rates were more likely to be classified into premium quality categories. Aung & Win (2022) studied rice quality evaluation in Myanmar using image processing techniques. The researchers found that rice samples with lower chalkiness rates were generally classified into higher quality grades. The study further suggested that chalkiness rate was an important quality indicator for evaluating Myanmar rice samples.

Similarly, Rahman et al. (2020) analyzed rice quality indicators using image-based measurements and observed that chalkiness rate significantly differed across rice quality categories. Rice samples with high chalkiness levels generally exhibited lower visual quality and poorer appearance characteristics. Hlaing et al. (2023) statistically analyzed the effect of chalkiness rate on rice grades using multinomial logistic regression analysis and reported that chalkiness rate significantly affected rice quality classification. The findings suggested that increases in chalkiness rate reduced the probability of rice samples being classified into higher quality categories. Previous empirical studies consistently indicated that chalkiness rate derived from digital image analysis provided objective, reliable, and statistically meaningful information for rice quality evaluation and automated rice grading systems (Zhang et al., 2021).

2.8 Yellow Rice Rate

Yellow rice rate referred to the proportion of yellow or discolored rice grains observed within rice samples (Juliano, 1993). Yellow rice defects were generally caused by improper storage conditions, fungal contamination, excessive moisture, aging, or unfavorable environmental factors during storage and processing (FAO, 2020). Rice grains affected by yellow discoloration were usually considered lower quality because discoloration negatively influenced rice appearance and consumer acceptance (Rahman et al., 2020). Previous studies reported that yellow rice defects significantly affected rice quality evaluation and rice grade classification systems (Verma, 2012). Yellow or discolored grains reduced the visual attractiveness and market value of rice samples because consumers generally preferred rice grains with natural and uniform color appearance (Li et al., 2019).

Silva & Sonnadara (2013) analyzed rice grain color characteristics using RGB intensity analysis and reported that yellow grains could be identified effectively using color-based image processing techniques. The researchers found that RGB color variations provided useful information for detecting discolored and defective rice grains automatically. Ramos et al. (2018) analyzed rice grading using geometric and photometric image descriptors and found that color related variables significantly influenced rice quality evaluation. Their study suggested that yellow rice defects negatively affected rice grade classification because rice samples containing higher proportions of yellow grains were more likely to be classified into lower quality categories.

Zhang et al. (2021) also investigated automated rice defect detection using image texture and color analysis methods. The researchers reported that color irregularities and yellow discoloration significantly contributed to automated rice defect detection systems. Their findings indicated that yellow rice grains produced noticeable variations in RGB intensity distributions and texture patterns. Similarly, Rahman et al. (2020) reported that rice samples with higher yellow rice rates generally exhibited poorer appearance quality and lower commercial value. The study concluded that yellow rice rate was an important defect related quality indicator in automated rice grading systems. Previous empirical studies consistently indicated that higher yellow rice rates were associated with lower quality rice grades and reduced market acceptance (Hlaing et al., 2023). Therefore, yellow rice rate was considered an important defect related indicator in digital image-based rice quality assessment systems.

2.9 Spot Rate

Spot rate referred to the proportion of rice grains containing visible spots, stains, or surface defects observed within rice samples (Juliano, 1993). Spot defects were generally caused by fungal infection, insect damage, improper storage conditions, or physical deterioration during rice processing and handling (FAO, 2020). Rice grains containing visible spots or stains were usually considered defective because such defects negatively affected rice appearance quality and market value (Rahman et al., 2020). Previous studies reported that spot defects significantly influenced rice quality evaluation and automated rice grading systems (Verma, 2012). Rice samples with higher spot rates generally exhibited poorer appearance quality and lower consumer acceptance because consumers preferred clean and visually uniform rice grains (Li et al., 2019).

Ponce-Garcia et al. (2017) investigated rice defect detection using image texture analysis techniques. The researchers analyzed spot defects using variance and texture related measurements and reported that spot defects could be identified effectively through digital image analysis methods. Their findings indicated that texture irregularities caused by spotted grains significantly affected image intensity distributions and surface characteristics. Zhang et al. (2021) developed automated rice defect detection models using machine learning and image analysis methods. The study reported that spot defects significantly affected rice appearance quality and classification performance. The researchers found that spotted grains produced

abnormal texture patterns and color variations that could be detected effectively using image processing techniques.

Similarly, Ramos et al. (2018) analyzed rice grading using geometric and photometric image descriptors and reported that defect related indicators such as spots and discoloration significantly influenced rice grade classification. Rice samples containing higher proportions of spotted grains were more likely to be classified into lower quality categories. Rahman et al. (2020) also compared rice quality indicators across different rice grades and observed that rice samples with higher spot rates generally exhibited poorer visual quality and lower market acceptance. The study concluded that spot rate was an important defect related quality indicator in rice grading systems. Previous empirical studies consistently indicated that spot rate derived from digital image analysis provided objective and statistically reliable information for rice quality evaluation and automated rice defect detection systems (Hlaing et al., 2023).

2.10 Rice Grade

Rice grade was considered the dependent variable in this study. Rice grades were commonly categorized into different quality classes based on rice appearance characteristics, grain uniformity, and defect levels (FAO, 2020). In rice grading systems, higher quality grades were generally assigned to rice samples with better appearance quality, fewer defects, lower chalkiness, and more uniform grain characteristics (Juliano, 1993). Previous studies reported that rice grades were significantly influenced by geometric indicators, photometric indicators, and defect related quality variables extracted from digital images (Ramos et al., 2018). Digital image processing techniques enabled researchers to classify rice samples objectively and consistently by analyzing measurable grain characteristics statistically (Verma, 2012).

Ramos et al. (2018) classified rice samples into different quality categories using geometric and photometric indicators derived from digital images. Their findings indicated that image-based indicators significantly influenced rice grade classification. The researchers reported that both grain shape characteristics and color related variables contributed to classification accuracy in automated rice grading systems. Rahman et al. (2020) compared rice quality indicators among different rice grades and found significant differences in geometric and color characteristics across rice quality categories. Higher quality rice grades generally exhibited more uniform grain

dimensions, lower defect rates, and better appearance characteristics compared with lower quality grades.

Li et al. (2019) also examined rice appearance quality using machine vision systems and reported that grain shape, color intensity, and chalkiness significantly influenced rice grade determination. The study concluded that image-based quality indicators improved the reliability and consistency of rice grading systems. Similarly, Zhang et al. (2021) developed automated rice defect detection and classification models using image processing and machine learning techniques. The researchers found that discoloration, surface defects, and texture irregularities significantly affected rice quality classification performance. Hlaing et al. (2023) applied multinomial logistic regression analysis to examine the effects of rice quality indicators on rice grades. Their study reported that aspect ratio, chalkiness rate, and RGB intensity values significantly affected rice grade classification. The findings suggested that both geometric and photometric indicators contributed significantly to the prediction of rice quality categories.

2.11 Review on Related Studies

Kuo et al. (2016) conducted a study on chalkiness detection in rice grains using digital image analysis techniques. Their study treated chalkiness rate as the primary dependent variable, while independent variables included grayscale intensity values and segmentation-based region properties extracted from rice grain images. Data were collected from rice grain samples using a flatbed scanner under controlled lighting conditions, with a total of 300 rice grains analyzed. Image preprocessing techniques including thresholding and morphological operations were applied to segment chalky regions from normal endosperm regions. The study employed statistical analysis of grayscale differences between chalky and normal regions to quantify chalkiness levels. The findings demonstrated that digital image processing methods effectively detected chalkiness defects and provided objective measurements of chalkiness severity. The researchers reported that brighter chalky regions produced higher grayscale intensity values compared to normal translucent regions, enabling accurate segmentation and classification of chalky grains.

Li et al. (2019) developed a machine vision system for rice appearance quality inspection, focusing on the extraction of geometric and photometric indicators for quality grading. Their study treated grain length, grain width, grain area, and chalkiness

degree as independent variables, while rice quality grade (premium, standard, and low-grade) served as the dependent variable. A total of 1,200 rice grain images were captured using a charge-coupled device (CCD) camera with uniform illumination conditions. The researchers applied image segmentation techniques including Otsu thresholding and watershed algorithm to isolate individual grains and extract geometric features. Multiple classification algorithms including support vector machine (SVM) and artificial neural network (ANN) were employed to classify rice into different quality grades. The findings revealed that geometric indicators, particularly grain length and aspect ratio, significantly influenced rice grade classification accuracy. The researchers reported that grain length was positively associated with higher quality grades, while chalkiness degree negatively affected quality classification, with classification accuracies exceeding 92% for premium-grade rice.

Silva & Sonnadara (2013) conducted a study on rice grain quality analysis using image processing techniques for color-based classification. The study treated RGB color intensities and grayscale features as independent variables, while rice quality categories (good, discolored, and defective) were treated as the dependent variable. A total of 500 rice grain images were acquired using a digital camera under standardized lighting conditions. Image preprocessing involved background removal, grayscale conversion, and color space transformation. The researchers employed statistical feature extraction methods to quantify RGB intensity distributions and grayscale mean values. Logistic regression and discriminant analysis were applied to classify rice grains based on extracted color features. The findings showed that RGB intensity values effectively distinguished between normal and discolored rice grains. Discolored grains exhibited lower red and green intensity values compared to normal grains, while yellow discoloration was associated with elevated red intensity levels relative to blue and green components. The study concluded that photometric indicators derived from digital images provided reliable measurements for automated rice quality classification.

Ponce-Garcia et al. (2017) investigated rice defect detection using image texture analysis techniques. The study treated variance and uniformity index as photometric indicators, along with geometric features such as grain area and perimeter, as independent variables, while defect categories (no defect, spot defect, chalky defect) were treated as the dependent variable. Rice grain images were captured using a flatbed scanner, with a dataset of 450 images comprising both defective and non-defective grains. Texture features were extracted using gray-level co-occurrence matrix (GLCM)

methods from grayscale-converted images. Statistical analysis including variance measurements and uniformity index calculations was applied to quantify intensity distributions within grain regions. The findings revealed that defective grains exhibited significantly higher intensity variance values compared to sound grains due to irregular surface texture patterns. The uniformity index was found to be lower for defective grains, indicating less consistent pixel intensity distributions across grain surfaces. The study reported that variance measurements contributed significantly to defect detection accuracy, with classification accuracies exceeding 85% for spot and chalkiness defects.

Ramos et al. (2018) analyzed rice grading using geometric and photometric image descriptors within a comprehensive quality classification framework. The study treated geometric indicators (grain length, width, area, aspect ratio) and photometric indicators (RGB intensities, grayscale mean, variance, uniformity index) as independent variables, while rice quality grade (Grade A, Grade B, and Grade C) served as the dependent variable. A total of 800 rice grain images were obtained from commercial rice samples representing three quality grades. Image processing involved segmentation, feature extraction, and data normalization procedures. The researchers employed multiple classification methods including linear discriminant analysis (LDA), k-nearest neighbors (K-NN), and support vector machines (SVM) to classify rice samples. The findings demonstrated that both geometric and photometric indicators significantly influenced rice grade classification. Aspect ratio was identified as the most discriminative geometric feature, while RGB intensity values contributed strongly to grade differentiation. The study reported that Grade A rice exhibited higher aspect ratio values (indicating slender grains) and more uniform color distributions compared to lower-grade samples.

Rahman et al. (2020) conducted a comparative analysis of rice quality indicators using image-based measurements across different quality grades. The study treated grain length, grain width, grain area, aspect ratio, and color intensity values as independent variables, while rice quality grade (premium, standard, and low) served as the dependent variable. A total of 600 rice grain images were captured from rice samples representing three distinct quality categories. Geometric features were extracted after image segmentation using morphological operations and thresholding techniques. Color features were extracted from RGB color space after illumination correction. Statistical analysis including analysis of variance (ANOVA) and correlation analysis was employed to compare quality indicators across rice grades. The findings revealed

significant differences in grain length, width, and aspect ratio across the three quality categories, with premium-grade rice exhibiting the longest and most slender grains. Color intensity values also differed significantly among grades, with premium-grade rice showing higher brightness and more uniform color characteristics. The study concluded that geometric and photometric indicators could reliably distinguish between different rice quality grades.

Hlaing et al. (2023) applied multinomial logistic regression analysis to examine the effects of rice quality indicators on rice grades, which is directly relevant to the methodological approach of the present study. The study treated aspect ratio, chalkiness rate, and RGB intensity values as independent variables, while rice grades (A, B, and C) served as the multinomial dependent variable. Data were collected from rice samples representing Myanmar rice varieties, with a total sample size of 400 rice grain images. Geometric and photometric features were extracted using digital image processing techniques, including grayscale conversion, thresholding, and morphological filtering. Multinomial logistic regression was employed to model the relationship between the quality indicators and the categorical outcome variable representing rice grade. The findings reported that aspect ratio, chalkiness rate, and RGB intensity values significantly affected rice grade classification. Specifically, higher aspect ratio values (slender grains) were associated with higher probability of Grade A classification, while elevated chalkiness rate increased the probability of Grade C classification. The study demonstrated that multinomial logistic regression provided an appropriate statistical framework for analyzing the effects of image-derived quality indicators on categorical rice grades.

Zhang et al. (2021) developed automated rice defect detection models using machine learning and image analysis methods. The study treated RGB intensity variations, grayscale mean values, and variance measurements as independent variables, while defect categories (no defect, yellow discoloration, spot defect, chalkiness) were treated as the dependent variable. A dataset of 1,500 rice grain images was acquired using a high-resolution digital camera under controlled lighting conditions. Feature extraction involved color space transformation, grayscale conversion, and texture analysis using GLCM methods. Machine learning models including logistic regression, random forest, and support vector machines were trained and validated for defect classification. The findings indicated that RGB intensity variations were strongly associated with rice discoloration and surface defects, with

yellow rice exhibiting distinct RGB ratios compared to sound rice. Variance measurements significantly improved defect detection accuracy, particularly for spot defects characterized by localized intensity variations. The logistic regression model achieved the highest classification accuracy at 94%, demonstrating the effectiveness of photometric indicators for automated rice quality assessment.

Asnidar et al. (2023) conducted a study on the classification of rice quality levels based on color and shape features using artificial neural networks based on digital image processing. The study treated color features and shape features (including grain length, width, and aspect ratio) as independent variables, while rice quality classes (good, good enough, and not good) served as the dependent variable. A total of 330 digital images were used in the study to produce a more accurate level of accuracy compared to previous studies with smaller datasets. The methodology included image retrieval, preprocessing, segmentation, morphological operations, feature extraction, and classification using artificial neural networks (JST). The findings reported a training accuracy of 98.75% and a test accuracy of 98.89%, demonstrating that color and shape features extracted through digital image processing could effectively classify rice into three distinct quality levels. The study concluded that the use of a larger dataset (330 images) significantly improved classification accuracy compared to previous studies using only 80 images.

Ruslan et al. (2024) presented a study on weedy rice classification using image processing and a machine learning approach, which is relevant to the broader context of rice quality assessment. The study treated morphological, color, and texture features extracted from RGB and monochrome images of seed kernels as independent variables, while the classification category (cultivated rice varieties versus weedy rice variant) served as the dependent variable. A machine vision unit was set up for image acquisition using an area scan camera, and a total of sixty-seven features were extracted from three primary parameters (morphology, color, and texture). Seven machine learning classifiers were used, including logistic regression (LR), and classification performance was evaluated using sensitivity, specificity, accuracy, and average correct classification metrics. The findings revealed that the best optimum model was developed using the RGB image with the logistic regression model, which achieved 85.3% sensitivity, 99.5% specificity, 97.9% accuracy, and 92.4% average correct classification utilizing all sixty-seven features. The study proved that features extracted from RGB images

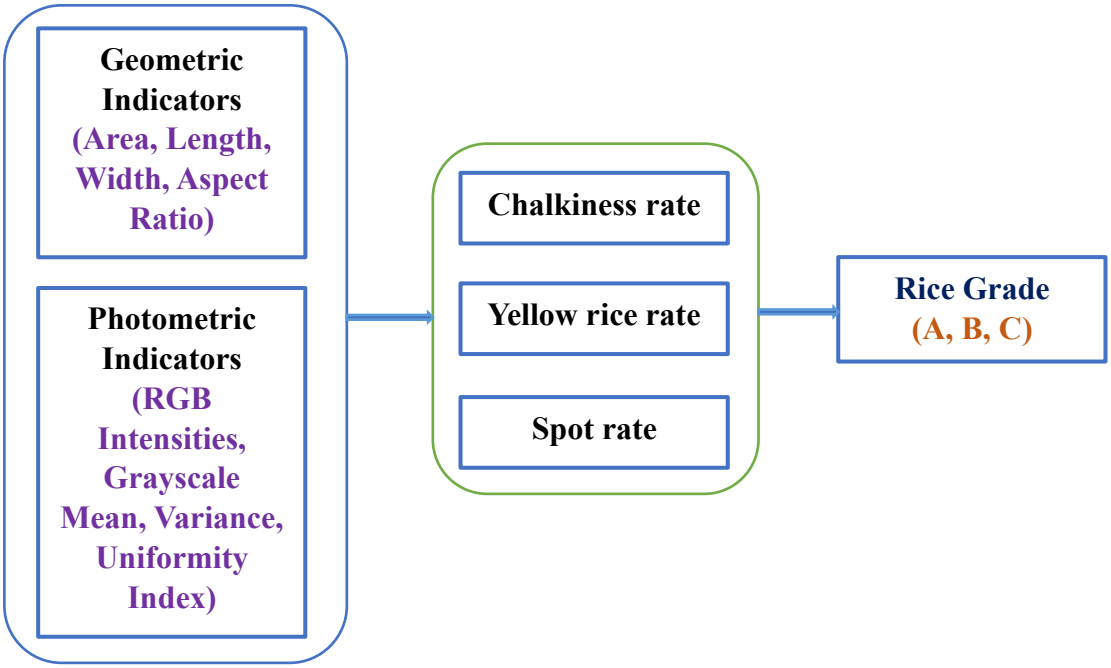
have higher sensitivity and accuracy in identifying rice seeds than monochrome images when using image processing and machine learning techniques.

2.12 Conceptual Framework of the Study

The conceptual framework of this study was developed based on previous empirical studies related to digital image processing and rice quality analysis. Variables used in the framework were adopted from earlier studies on rice quality indicators derived from digital images.

In this study, geometric indicators (area, length, width, and aspect ratio), photometric indicators (RGB intensities, grayscale mean, variance, and uniformity index), and defect related indicators (chalkiness rate, yellow rice rate, and spot rate) were treated as independent variables. Rice grade categorized as Grade A, Grade B, and Grade C was treated as the dependent variable. Previous studies reported that these rice quality indicators significantly influenced rice grade classification (Sakulsingharoj et al., 2010; Ramos et al., 2018; Rahman et al., 2020; Hlaing et al., 2023). Therefore, the conceptual framework presented in Figure (2.1) was developed by the researcher based on the reviewed literature.

Figure (2.1) Conceptual Framework of the Study



Source: Own Compilation

CHAPTER III
RESEARCH METHODOLOGY

This chapter describes the research methodology used in the study. It includes the research design, population and sampling procedures, data collection methods, image acquisition and preprocessing techniques, feature extraction procedures, variable measurements, and statistical analysis methods employed in the study. In addition, the assumptions related to the statistical methods used in the analysis are also presented.

3.1 Study Area and Design

This study adopted a quantitative research design to analyze rice quality indicators derived from digital image measurements. The study focused on three rice grade categories: Grade A, Grade B, and Grade C. Rice samples were collected from local rice markets and rice distribution sources in Yangon Region, Myanmar. Primary data were used in the study, and the data were obtained directly from digital images of rice grain samples.

A two-stage cluster sampling method was applied in the sampling procedure. In the first stage, a large rice sack was treated as the population cluster, and a “pyay tone” (traditional sampling bag) was selected as the first cluster sample. In the second stage, a small cup was used to draw the final rice grain subsample from the selected cluster. The study consisted of 100 digital images for each rice grade category, resulting in a total sample size of 300 images. The collected rice grains were scanned using a flatbed scanner under controlled lighting and background conditions to obtain digital images for subsequent image processing procedures.

3.2 Sample Size Determination

The sample size of the study was determined based on the number of rice grain images collected for rice quality analysis and rice grade classification. The study focused on three rice grade categories: Grade A, Grade B, and Grade C. To ensure adequate representation and sufficient observations for statistical analysis, 100 digital images were collected from each rice grade category. Therefore, the total sample size of the study consisted of 300 rice grain images.

Each digital image represented a batch rice sample containing approximately 300 to 900 rice grains depending on grain size and grain arrangement during image

acquisition. The collected images were treated as the primary observational units for image processing and statistical analysis.

The sample size of the study can be expressed as:

$$n = n_A + n_B + n_C$$

where:

- $n$represented the total sample size
- $n_A$represented the number of images for Grade A rice
- $n_B$represented the number of images for Grade B rice
- $n_C$represented the number of images for Grade C rice

Substituting the sample sizes for each rice grade category:

$$n = 100 + 100 + 100 = 300$$

Thus, the total sample size of the study consisted of 300 digital rice grain images.

3.3 Digital Image Processing and Feature Extraction

Digital image processing techniques were employed in this study to obtain objective and quantitative measurements of rice quality indicators from rice grain images. Image processing methods provide efficient and consistent procedures for analyzing rice grain characteristics compared with traditional manual inspection methods (Gonzalez & Woods, 2018). In this study, digital image processing procedures were performed sequentially through image acquisition, image preprocessing, image segmentation, and feature extraction processes. These procedures enabled the conversion of rice grain images into measurable variables suitable for statistical analysis and rice grade classification.

Let the digital image be represented by:

$$I(x, y), x = 1,2, \dots, w; y = 1,2, \dots, h$$

where:

- w= image width
- h= image height
- $I(x, y)$= pixel intensity value at coordinate (x, y)

The collected images were RGB color images consisting of red, green, and blue color channels.

3.3.1 Image Acquisition

Image acquisition was the first stage of the digital image processing procedure. Rice grain samples were scanned using a flatbed scanner under controlled lighting conditions to obtain high quality digital images. Uniform background and lighting conditions were maintained to reduce shadows, reflections, and image inconsistencies during image capture (Verma, 2012).

The captured images were stored in RGB color format because RGB channels provide important photometric information related to rice grain color and appearance characteristics (Silva & Sonnadara, 2013). Proper image acquisition was essential because image quality directly influenced the accuracy of image preprocessing, segmentation, and feature extraction procedures.

According to Gonzalez and Woods (2018), digital images can be represented mathematically as two-dimensional functions:

$$f(x, y)$$

where $x$and $y$represent spatial coordinates of the image and $f(x, y)$represents the intensity value at pixel location (x, y). The RGB image can also be represented as:

$$I(x, y) = \{R(x, y),\ G(x, y),\ B(x, y)\}$$

where:

- $R(x, y)$= red channel intensity
- $G(x, y)$= green channel intensity
- $B(x, y)$= blue channel intensity

3.3.2 Image Preprocessing

Image preprocessing techniques were applied to improve image quality before segmentation and feature extraction. Preprocessing procedures help reduce noise, improve contrast, and standardize image characteristics for subsequent analysis (Gonzalez & Woods, 2018). The preprocessing procedures employed in this study included image normalization, noise reduction, grayscale conversion, and thresholding.

(a) Image Normalization (Z-score Normalization)

Image normalization was applied to reduce variations caused by illumination intensity and image acquisition conditions. In this study, Z-score normalization was employed to standardize pixel intensity values across images, ensuring that all images had similar intensity distributions before further processing.

The normalization process was expressed as:

$$I' = \frac{I - \mu}{\sigma}$$

where:

- I'= normalized image intensity
- I= original image intensity
- μ= mean intensity value of the image
- σ= standard deviation of the image intensities

After normalization, the transformed image data had mean zero and variance one, which helps reduce bias due to lighting differences between images.

(b) Noise Reduction (Median Filtering)

Noise reduction techniques were applied to remove unwanted disturbances from the images while preserving grain boundaries. Median filtering was employed because it effectively removes salt-and-pepper noise commonly found in digital images (Verma, 2012).

The median filtering operation can be expressed as:

$$I'(m,n) = \text{median}\{\, I(i,j) : (i,j) \in w \,\}$$

where:

- $I'(m,n)$= filtered pixel value at coordinates (m, n)
- $I(i,j)$= neighboring pixel values
- w= filtering window (e.g., 3×3)

The operation follows these steps: a window of pixels surrounding the target pixel is selected, the pixel values within that window are sorted in ascending order, and the middle (median) value is chosen as the new value for the target pixel. This technique removes impulsive noise while keeping edges sharp.

(c) Grayscale Conversion

RGB images were converted into grayscale images to simplify image analysis and reduce computational complexity. The grayscale conversion formula was expressed as:

$$I_{gray} = 0.299R + 0.587G + 0.114B$$

where:

- R, G, B= red, green, and blue intensity values, respectively
- I_{gray}= resulting grayscale intensity value

This weighted conversion method is commonly used in image processing studies because it reflects human visual sensitivity to different color channels (Gonzalez & Woods, 2018).

(d) Thresholding

Thresholding techniques were applied to separate rice grains from the background. Binary image conversion was performed by assigning white pixels to rice grains and black pixels to the background.

The thresholding function was expressed as:

$$g(x,y) = \begin{cases} 1, & f(x,y) \geq T \\ 0, & f(x,y) < T \end{cases}$$

where:

- T= threshold value
- $g(x,y)$= binary image output

Thresholding procedures improve segmentation accuracy by enhancing object boundary detection (Ramos et al., 2018).

3.3.3 Image Segmentation

Image segmentation was performed to isolate individual rice grains from the background. Accurate segmentation was important because segmentation errors could affect subsequent feature extraction results (Sakulsingharoj et al., 2010).

Connected component labeling techniques were employed to identify individual rice grains. The segmentation process used 8-neighborhood connectivity analysis to determine connected pixels belonging to the same rice grain object.

The 8-neighborhood system was defined as:

$$N_8(x,y) = \{(x \pm 1, y),\ (x, y \pm 1),\ (x \pm 1, y \pm 1)\}$$

Connected component labeling was then applied using:

$$L(x,y) = \begin{cases} i, & (x,y) \in C_i \\ 0, & (x,y) \in \text{Background} \end{cases}$$

where:

- C_i= i-th connected component
- $L(x,y)$= assigned label of each pixel

After segmentation, each rice grain was isolated as an individual object for feature extraction analysis.

3.3.4 Feature Extraction

Feature extraction was performed after image segmentation to obtain measurable rice quality indicators from segmented rice grain images. According to Gonzalez and Woods (2018), feature extraction refers to the process of converting image information into quantitative variables suitable for computational and statistical analysis. In rice quality analysis, extracted image features are commonly categorized into geometric features, photometric features, and defect-related features (Ramos et al., 2018).

In this study, geometric indicators including area, length, width, and aspect ratio; photometric indicators including RGB intensities, grayscale mean, variance, and uniformity index; and defect-related indicators including chalkiness rate, yellow rice rate, and spot rate were extracted from rice grain images.

(a) Area

Area was measured to represent the total pixel region occupied by a rice grain in the segmented binary image. Previous studies used grain area measurements to evaluate grain size characteristics and rice quality classification (Sakulsingharoj et al., 2010; Verma, 2012).

The area measurement was calculated as:

$$A_i = \sum_{(x,y)} [L(x,y) = i]$$

where:

- A_i= area of the i-th rice grain
- $L(x,y)$= labeled segmented image

(b) Length and Width

Length was extracted to measure the major axis dimension of rice grains. Grain length measurements were commonly employed in rice appearance quality studies and rice classification systems (Li et al., 2019). Width represented the minor axis dimension of rice grains measured perpendicular to grain length. Width measurements were frequently analyzed together with grain length in rice grain morphology studies (Verma, 2012).

The grain length and width were expressed as:

$$\text{Length} = \max(d_{ij}), \text{Width} = \min(d_{ij})$$

where d_{ij}represents the Euclidean distance between boundary points of the rice grain.

(c) Aspect Ratio

Aspect ratio was calculated to measure grain shape characteristics. Previous studies reported that aspect ratio was an important geometric indicator in rice quality classification systems (Sakulsingharoj et al., 2010; Rahman et al., 2020).

$$\text{Aspect Ratio} = \frac{\text{Length}}{\text{Width}}$$

Higher aspect ratio values generally indicated more slender rice grains.

(d) RGB Mean Intensities

RGB intensity values were extracted to represent color information from rice grain images. According to Silva and Sonnadara (2013), RGB color analysis was commonly used in rice appearance evaluation and defect detection studies.

The RGB mean intensity value was calculated as:

$$\mu_c = \frac{1}{N} \sum_{(x,y) \in C_i} I_c\,(x, y)$$

where:

- μ_c= mean intensity of color channel c(R, G, or B)
- $I_c(x, y)$= pixel intensity value for channel c
- N= total number of pixels within the grain region

The extracted variables included R_{mean}, G_{mean}, and B_{mean}.

(e) Grayscale Mean

Grayscale mean was extracted to represent the average grayscale intensity value of rice grain images after grayscale conversion. According to Kuo et al. (2016), grayscale intensity analysis was commonly applied in rice image processing studies because it simplified image representation while preserving brightness information required for texture and defect analysis.

The grayscale mean was calculated as:

$$\text{Gray Mean} = \frac{1}{N} \sum_{i=1}^{N} g_i$$

where:

- g_i= grayscale intensity values of pixels within the grain
- N= total number of pixels within the rice grain region

(f) Variance

Variance was extracted to measure the variability of grayscale intensity values within rice grain images. Previous studies reported that variance measurements were useful for texture analysis and defect detection in agricultural image processing applications (Ponce-Garcia et al., 2017).

The variance was calculated as:

$$\sigma^2 = \sum_{i=0}^{L-1} (z_i - m)^2 p(z_i)$$

where:

- z_i= grayscale intensity level
- m= mean grayscale intensity
- $p(z_i)$= probability distribution of intensity level z_i
- L= total number of grayscale intensity levels

Higher variance values indicated greater texture irregularity and intensity variation.

(g) Uniformity Index

Uniformity index was extracted to measure the consistency of grayscale intensity distribution within rice grain images. According to Ramos et al. (2018), uniformity measurements were commonly employed in image texture analysis because they represented the degree of homogeneity within image regions.

The uniformity index was calculated as:

$$U = \sum_{i=0}^{L-1} p^2 (z_i)$$

where:

- $p(z_i)$= probability of grayscale intensity level z_i
- L= total number of grayscale intensity levels

Higher uniformity values indicated more homogeneous grain surfaces.

(h) Chalkiness Rate

Chalkiness rate was measured to represent the proportion of chalky rice grains within each image. Previous studies identified chalkiness as an important rice appearance defect because chalky grains affected rice appearance quality and milling performance (Kuo et al., 2016; Li et al., 2019).

The chalkiness rate was calculated as:

$$CR = \frac{n_{\text{chalky}}}{n} \times 100\%$$

where:

- n_{chalky}= number of chalky rice grains
- n= total number of rice grains in the image

(i) Yellow Rice Rate

Yellow rice rate was measured to represent the proportion of yellow or discolored rice grains within each image. Silva and Sonnadara (2013) reported that yellow discoloration could be identified effectively using image-based color analysis techniques.

$$YR = \frac{n_{\text{yellow}}}{n} \times 100\%$$

where:

- n_{yellow}= number of yellow rice grains
- n= total number of rice grains

(j) Spot Rate

Spot rate was measured to represent the proportion of rice grains containing visible spots or stains. Previous studies reported that spot defects were important indicators in rice appearance quality evaluation (Zhang et al., 2021).

$$SR = \frac{n_{\text{spot}}}{n} \times 100\%$$

where:

- n_{spot}= number of rice grains containing visible spots
- n= total number of rice grains

3.3.5 Data Aggregation to Image Level

Since each image contained multiple rice grains (approximately 300–900 grains), the grain-level measurements described in Section 3.3.4 were aggregated to the image level to create independent observations for statistical analysis. For each image i, the following image-level variables were computed:

- Geometric indicators: average grain length, average grain width, average aspect ratio
- Photometric indicators: average RGB mean intensities, average grayscale mean, average variance, average uniformity index

- Defect-related indicators: chalkiness rate, yellow rice rate, spot rate (already computed as proportions at image level), broken rice rate, etc.

  The aggregation for an average (e.g., average grain length) was performed as:

$$\bar{L}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} L_{ij}$$

where:

- $\bar{L}_i$= image-level average length for image i
- n_i= number of grains in image i
- L_{ij}= length of the j-th grain in image i

Similar averages were computed for width, aspect ratio, and photometric indicators. Defect rates were calculated as the proportion of affected grains within the image, as shown in Sections 3.3.4 (h)–(j). This aggregation procedure produced a final dataset with 300 rows (one per image), where each row contained the image-level variables. The aggregated dataset satisfied the independence assumption required for classical statistical inference because each image was treated as a single, independent observation.

3.4 Descriptive Statistics

Descriptive statistics were employed to summarize and describe the basic characteristics of the observed data. According to Walpole et al. (2012), descriptive statistics are statistical methods used to organize, summarize, and present data in meaningful forms. Descriptive measures help researchers understand the distribution, central tendency, and variability of variables before conducting inferential statistical analysis.

(a) Mean

The arithmetic mean was used to measure the average value of observations. Montgomery and Runger (2014) stated that the mean is one of the most widely used measures of central tendency in statistical analysis.

The sample mean was calculated as:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

where:

- $\bar{X}$represented the sample mean

- $X_i$represented the observed value of the i^{th}observation
- $n$represented the total number of observations

(b) Standard Deviation

Standard deviation was used to measure the variability or dispersion of observations around the sample mean. According to Walpole et al. (2012), standard deviation indicates the degree to which observations deviate from the mean value.

The sample standard deviation was calculated as:

$$S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$$

where:

- Srepresented the sample standard deviation
- $X_i$represented observed values
- $\bar{X}$represented the sample mean
- $n$represented the sample size

(c) Minimum and Maximum Values

Minimum and maximum values were used to identify the smallest and largest observations in the dataset. These measures provided information regarding the range and spread of the observed data (Montgomery & Runger, 2014).

The minimum and maximum values were expressed as:

$$X_{min} = min(X_i)$$
$$X_{max} = max(X_i)$$

where:

- X_{min}represented the minimum observed value
- X_{max}represented the maximum observed value

(d) Coefficient of Variation

The coefficient of variation (CV) was used to measure relative variability among variables. According to Walpole et al. (2012), the coefficient of variation expresses the standard deviation relative to the mean and is useful for comparing variability among variables measured on different scales.

The coefficient of variation was calculated as:

$$CV = \frac{S}{\bar{X}} \times 100\%$$

where:

- CVrepresented the coefficient of variation
- Srepresented the standard deviation
- $\bar{X}$represented the sample mean

The coefficient of variation was expressed as a percentage.

3.5 One-way Analysis of Variance (ANOVA)

One-way Analysis of Variance (ANOVA) was employed to examine whether significant mean differences existed among groups. According to Montgomery and Runger (2014), one-way ANOVA is an inferential statistical method used to compare the means of three or more independent groups simultaneously. ANOVA helps determine whether observed differences among group means are statistically significant. One-way ANOVA was applied to compare the mean values of variables across the three groups.

The hypotheses for one-way ANOVA were expressed as:

$$H_0: \mu_1 = \mu_2 = \mu_3$$

H_1:At least one group mean differs

where:

- H_0represented the null hypothesis
- H_1represented the alternative hypothesis
- μ_1, μ_2, μ_3represented the population means of the groups

The ANOVA F statistic was calculated as:

$$F = \frac{MSB}{MSW}$$

where:

- MSBrepresented the mean square between groups
- MSWrepresented the mean square within groups

The mean square between groups was calculated as:

$$MSB = \frac{SSB}{k-1}$$

The mean square within groups was calculated as:

$$MSW = \frac{SSW}{N-k}$$

where:

- SSBrepresented the sum of squares between groups
- SSWrepresented the sum of squares within groups
- $k$represented the number of groups
- Nrepresented the total number of observations

The decision rule for the ANOVA test was based on the p-value at the 5 percent significance level:

- If $p < 0.05$, the null hypothesis was rejected.
- If $p \geq 0.05$, the null hypothesis was not rejected.

According to Field (2018), several assumptions should be satisfied before conducting one-way ANOVA.

(a) Independence Assumption

The observations were assumed to be independent of one another. Independence was maintained through the data collection and sampling procedures.

(b) Normality Assumption

The normality assumption stated that the data within each group should be approximately normally distributed. According to Shapiro and Wilk (1965), the Shapiro-Wilk test is commonly used to examine normality.

The Shapiro-Wilk test statistic was expressed as:

$$W = \frac{(\sum_{i=1}^{n} a_i \, x_i)^2}{\sum_{i-1}^{n} (x_i - \bar{x})^2}$$

where:

- Wrepresented the Shapiro-Wilk test statistic
- $a_i$represented constants generated from expected normal order statistics
- $x_i$represented ordered sample observations
- $\bar{x}$represented the sample mean

The decision rule for the Shapiro-Wilk test was:

- If $p > 0.05$, the normality assumption was satisfied.
- If $p \leq 0.05$, the normality assumption was violated.

In addition, histogram plots and Q-Q plots were examined to assess the normality of the data distribution.

(c) Homogeneity of Variance Assumption

The homogeneity of variance assumption stated that group variances should be approximately equal. According to Levene (1960), Levene's test is commonly used to assess equality of variances across groups.

The Levene's test statistic was expressed as:

$$W = \frac{\left(\sum_{i=1}^{k} n_i \left(Z_{i.} - Z_{..}\right)^2\right)}{\left(\sum_{i=1}^{k} \sum_{j=1}^{n_i} (Z_{ij} - Z_{i.}\right)^2}$$

where:

- Nrepresented the total sample size
- $k$represented the number of groups
- $n_i$represented the sample size of group i
- Z_{ij}represented transformed observations

The decision rule for Levene's test was:

- If $p > 0.05$, the homogeneity assumption was satisfied.
- If $p \leq 0.05$, the homogeneity assumption was violated.

(d) Post Hoc Test

When the result of the one-way ANOVA was statistically significant ($p < 0.05$), post hoc multiple comparison tests were conducted to identify which group means differed significantly from one another. Although the ANOVA F-test indicated the existence of significant mean differences among groups, it did not specify the particular pairs of groups responsible for the differences (Field, 2018). Therefore, additional pairwise comparison procedures were performed following a significant ANOVA result.

Post hoc tests were selected based on the result of the homogeneity of variance assumption assessed by Levene's test.

- If the homogeneity of variance assumption was satisfied ($p > 0.05$), Tukey's Honestly Significant Difference (HSD) test was applied. Tukey's HSD test is appropriate when group variances are equal and is commonly used to control the family-wise Type I error rate in multiple comparisons (Montgomery & Runger, 2014).
- If the homogeneity of variance assumption was violated ($p \leq 0.05$), the Games–Howell test was employed because it does not require equal group

variances and provides more reliable pairwise comparisons under heteroscedasticity (Field, 2018).

The Tukey HSD statistic was calculated as:

$$HSD = q\sqrt{\frac{MSW}{n}}$$

where:

- $q$represented the studentized range statistic
- MSWrepresented the mean square within groups obtained from the ANOVA table
- $n$represented the sample size per group

Pairwise comparisons were conducted among all group combinations to identify significant differences between group means. Statistical significance for post hoc comparisons was evaluated at the 5 percent significance level ($alpha = 0.05$).

If the adjusted p-value for a pairwise comparison was less than 0.05, the difference between the corresponding group means was considered statistically significant. Conversely, if the adjusted p-value was greater than or equal to 0.05, the difference was considered not statistically significant.

3.6 Pearson's Correlation Analysis

Pearson's correlation analysis was employed to examine the strength and direction of linear relationships between variables. According to Walpole et al. (2012), correlation analysis is a statistical technique used to measure the degree of association between two quantitative variables. Pearson's product moment correlation coefficient is one of the most widely used correlation measures in statistical analysis.

The hypotheses for Pearson's correlation analysis were expressed as:

$$H_0: \rho = 0$$

$$H_1: \rho \neq 0$$

where:

- H_0represented the null hypothesis indicating no linear relationship between variables
- H_1represented the alternative hypothesis indicating a significant linear relationship between variables
- ρrepresented the population correlation coefficient

The Pearson correlation coefficient was calculated as:

$$r = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2 \sum_{i=1}^{n}(Y_i - \bar{Y})^2}}$$

where:

- $r$represented the Pearson correlation coefficient
- $X_i$and $Y_i$represented observed values of variables $X$and Y
- $\bar{X}$and $\bar{Y}$represented sample means
- $n$represented the number of observations

The value of the Pearson correlation coefficient ranged from $-1$to $+1$.

- $r = +1$indicated a perfect positive linear relationship
- $r = -1$indicated a perfect negative linear relationship
- $r = 0$indicated no linear relationship

According to Evans (1996), the strength of correlation coefficients may be interpreted as follows:

Correlation Coefficient r	**Interpretation**
0.00 – 0.19	Very weak
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very strong

The significance of the correlation coefficient was tested using the t statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

where:

- $t$represented the test statistic
- $r$represented the Pearson correlation coefficient
- $n$represented the sample size

The decision rule for the significance test was based on the p-value at the 5 percent significance level:

- If $p < 0.05$, the null hypothesis was rejected.
- If $p \geq 0.05$, the null hypothesis was not rejected.

According to Field (2018), Pearson's correlation analysis assumes that the variables are measured at the interval or ratio scale, approximately normally distributed, linearly related, and free from extreme outliers. In this study, scatter plots were examined to assess linearity, while histograms, Q-Q plots, and the Shapiro-Wilk test were used to assess normality. Boxplots were also examined to identify potential outliers before conducting the correlation analysis.

3.7 Multinomial Logistic Regression Analysis

Multinomial logistic regression analysis was employed to examine the effects of rice quality indicators on rice grade classification. The dependent variable consisted of three unordered categorical outcomes: Grade A, Grade B, and Grade C. According to Hosmer et al. (2013), multinomial logistic regression is appropriate when the dependent variable contains more than two nominal categories. The method estimates the probability of membership in each outcome category relative to a selected reference category.

In this study, Grade C was selected as the reference category. Therefore, the model estimated the probabilities of rice samples being classified into Grade A or Grade B relative to Grade C.

Let Ydenote the dependent variable with $J = 3$outcome categories. The multinomial logistic regression model was expressed as:

$$\ln\left(\frac{P(Y=j)}{P(Y=J)}\right) = \beta_{0j} + \beta_{1j}X_1 + \beta_{2j}X_2 + \cdots + \beta_{kj}X_k$$

for $j = 1,2$, where:

- $P(Y=j)$represented the probability of belonging to category j
- $P(Y=J)$represented the probability of belonging to the reference category
- β_{0j}represented the intercept parameter
- β_{kj}represented the regression coefficient of the k^{th}predictor
- $X_1, X_2, \ldots, X_k$represented the independent variables

The independent variables included geometric indicators, photometric indicators, and defect related indicators extracted from digital rice grain images.

The regression coefficients were estimated using the Maximum Likelihood Estimation (MLE) method. According to Agresti (2018), MLE determines parameter estimates that maximize the likelihood of observing the sample data.

The likelihood function was expressed as:

$$L(\beta) = \prod_{i=1}^{n} \prod_{j=1}^{J} \pi_{ij}^{y_{ij}}$$

where:

- $L(\beta)$represented the likelihood function
- π_{ij}represented the predicted probability for observation $i$in category j
- y_{ij}represented the observed outcome value

3.7.1 Significance Testing of Regression Coefficients

The significance of individual regression coefficients was evaluated using the Wald chi-square statistic (Hosmer et al., 2013). The Wald statistic tested whether a regression coefficient differed significantly from zero.

The Wald statistic was calculated as:

$$W = \left(\frac{\hat{\beta}}{SE(\hat{\beta})}\right)^2$$

where:

- $\hat{\beta}$represented the estimated regression coefficient
- $SE(\hat{\beta})$represented the standard error of the coefficient estimate

If the p-value associated with the Wald statistic was less than 0.05, the corresponding independent variable was considered statistically significant.

The overall significance of the multinomial logistic regression model was evaluated using the Likelihood Ratio Test. The likelihood ratio statistic was expressed as:

$$G^2 = -2\ln\left(\frac{L_0}{L_1}\right)$$

where:

- L_0represented the likelihood of the reduced model
- L_1represented the likelihood of the full model

A statistically significant likelihood ratio test indicated that the fitted model provided better explanatory power than the intercept-only model.

3.7.2 Model Fit Evaluation

Pseudo coefficient of determination measures was used to evaluate model fit, including Cox and Snell R^2, Nagelkerke R^2, and McFadden R^2. According to Menard (2010), pseudo R^2measures provide approximate indications of explanatory power in logistic regression models.

In addition, classification accuracy was evaluated using a classification table (confusion matrix). Predicted rice grades obtained from the fitted model were compared with observed rice grades to assess the percentage of correctly classified observations.

3.7.3 Interpretation of Odds Ratios

The effects of independent variables were interpreted using odds ratios. The odds ratio for each predictor variable was calculated as:

$$OR = e^{\beta}$$

where:

- ORrepresented the odds ratio
- βrepresented the estimated regression coefficient

An odds ratio greater than one indicated that an increase in the predictor variable increased the odds of belonging to the comparison category relative to the reference category. Conversely, an odds ratio less than one indicated that an increase in the predictor variable decreased the odds of belonging to the comparison category.

3.7.4 Assumptions of Multinomial Logistic Regression

Assumptions were considered before conducting multinomial logistic regression analysis (Hosmer et al., 2013; Field, 2018).

(a) Independence of Observations

The observations were assumed to be independent. Each rice grain image was treated as an independent observation during data collection and statistical analysis.

(b) Multicollinearity

Multicollinearity refers to high correlations among independent variables. Severe multicollinearity may produce unstable regression coefficient estimates and inflated standard errors (Gujarati & Porter, 2009).

Multicollinearity was assessed using the Variance Inflation Factor (VIF), calculated as:

$$VIF = \frac{1}{1 - R_j^2}$$

where R_j^2represented the coefficient of determination obtained by regressing the j^{th}independent variable on the remaining predictors.

The following criteria were used:

- $VIF < 10$: no serious multicollinearity
- $VIF \geq 10$: potential multicollinearity problem

(c) Linearity of the Logit

For continuous independent variables, multinomial logistic regression assumes a linear relationship between continuous predictors and the logit of the outcome variable. This assumption was evaluated using the Box–Tidwell procedure (Hosmer et al., 2013).

If the interaction terms between predictors and their logarithmic transformations were not statistically significant ($p > 0.05$), the linearity assumption was considered satisfied.

(d) Absence of Extreme Outliers

Extreme outliers may influence regression estimates and reduce model stability. Standardized residuals and boxplots were examined to identify potential outliers before model estimation.

Observations with standardized residual values exceeding ± 2.5were treated as potential outliers and were carefully reviewed.

(e) Sample Size Adequacy

Adequate sample size is important for obtaining stable multinomial logistic regression estimates (Hosmer et al., 2013). The study included 300 observations distributed equally across the three rice grade categories, which provided sufficient observations for model estimation and classification analysis.

The significance level for all statistical tests in the multinomial logistic regression analysis was set at the 5 percent level (α = 0.05).

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